

E-Commerce Sales & Profitability Analysis

Exploratory Data Analysis Project Report (Python)

Dataset: 3,200 orders | Jan 2024 – Dec 2025 | India

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn

Table of Contents

1. Executive Summary

This report presents an end-to-end exploratory data analysis (EDA) of an e-commerce transactions dataset containing 3,200 orders placed between 1 January 2024 and 31 December 2025 across 12 Indian cities. The analysis covers data quality, cleaning, business KPIs, product, time, customer, geographic, channel and discount analysis, and RFM customer segmentation.

Headline findings (realised revenue = delivered orders only):

- Realised revenue of ₹52.52M and profit of ₹10.48M (19.96% margin) from 2,933 orders and 823 customers.
- Electronics generates 72% of revenue but has the lowest margin (15.1%); Laptop Pro 14 alone is 47% of revenue.
- Revenue is flat year over year (–1.4% in 2025); Q4 and November are seasonal peaks.
- Deeper discounts reduce margin from 38.6% (no discount) to 22.2% (16–25% discount).
- 8.3% of orders are returned or cancelled, equal to ₹5.06M of gross sales.
- The customer base is loyal (88% repeat buyers) but 47% of buyers are At Risk or Lost by RFM.

2. Objectives & Scope

- Assess overall business performance (revenue, profit, margin, order value).
- Identify best and worst performing categories, products, regions and channels.
- Understand time trends and seasonality.
- Segment customers with RFM and identify value concentration.
- Quantify the effect of discounts on profitability and provide recommendations.

3. Dataset Description

The raw file contains 3200 rows and 22 columns. Each row is one order line with a single product. Dates span 2024-01-01 to 2025-12-31.

Group	Columns
Order	Order_ID, Order_Date, Order_Status
Customer	Customer_ID, Customer_Age (18–65), Customer_Gender
Geography	City (12), State (11), Region (5)
Product	Product_ID, Product_Name (24), Category (6)
Financial	Quantity, Unit_Price, Discount_Pct, Sales, Cost, Profit
Operations	Payment_Method, Sales_Channel, Shipping_Type, Customer_Rating

4. Data Quality & Cleaning

4.1 Audit

Check	Result
Shape	3,200 rows × 22 columns
Duplicate rows	0
Missing – Customer_Gender	25 (0.8%)

Check	Result
Missing – Discount_Pct	32 (1.0%)
Missing – Customer_Rating	48 (1.5%)
Profit = Sales – Cost	True for all rows
Sales = Qty × Price × (1 – Discount)	99.2% of rows
Orders with negative profit	15

4.2 Cleaning decisions

- Gender nulls filled with "Unknown" so no rows are lost.
- Discount nulls filled with 0 (assumption: no discount applied). The mismatch rows in the Sales formula check are consistent with these 32 records.
- Rating nulls kept as NaN and flagged (Rating_Missing) so the column stays numeric for averages and correlations.
- Columns renamed (Age, Gender, Discount, Product_Category); Year, Month, Quarter, Age_Group, Margin_Pct and discount bands engineered.

4.3 Analysis basis

Because 168 orders were returned and 99 cancelled, revenue KPIs are reported on delivered orders (realised revenue). Gross figures including all statuses are shown for reference.

5. Overall Business Performance

KPI	Delivered only	All orders
Revenue	₹52.52M	₹57.58M
Cost	₹42.04M	₹46.17M
Profit	₹10.48M	₹11.41M
Profit margin	19.96%	19.81%
Units sold	5,430	5,899
Orders	2,933	3,200
Customers	823	832
Average order value	₹17,906	₹17,995

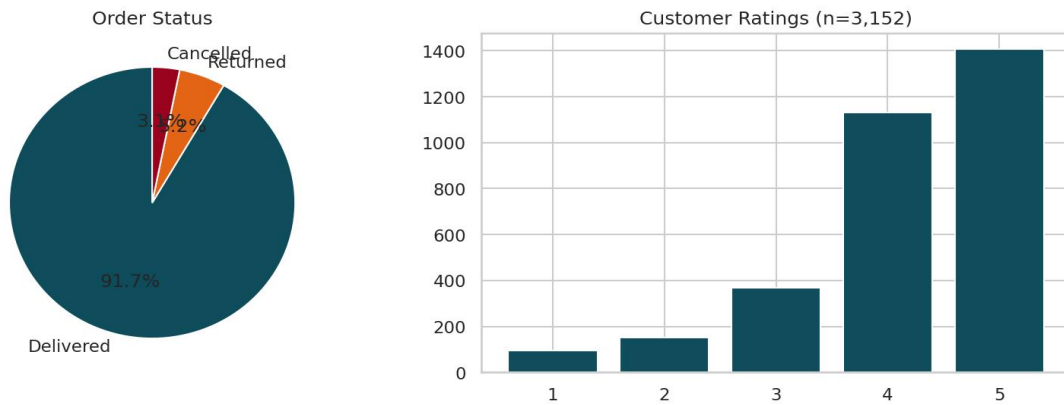


Figure 1: Order status split and customer rating distribution

Returned and cancelled orders represent 8.3% of orders and ₹5.06M of gross sales (Returned ₹3.52M, Cancelled ₹1.54M). The average customer rating is 4.14/5; 44.6% of rated orders are 5-star and 7.9% are 1–2 star.

6. Product & Category Analysis

Product_Category	Revenue	Profit	Orders	Margin
Electronics	₹37.75M	₹5.70M	835	15.11%
Furniture	₹5.94M	₹1.73M	442	29.1%
Home & Kitchen	₹3.80M	₹1.17M	461	30.82%
Fashion	₹2.24M	₹0.80M	520	35.61%
Sports	₹1.67M	₹0.59M	318	35.13%
Beauty	₹1.13M	₹0.50M	357	44.26%

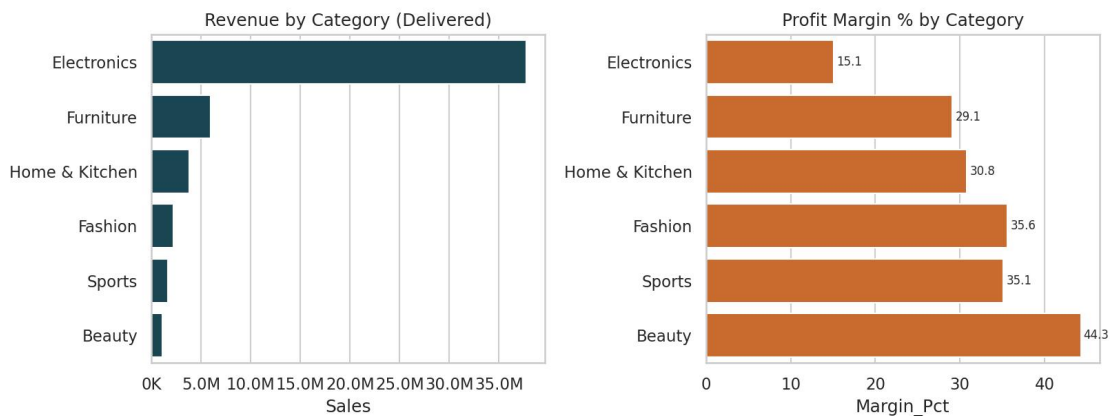


Figure 2: Revenue and profit margin by category

Electronics contributes ₹37.7M (72%) of revenue but converts only 15.1% into profit, while Beauty earns 44.3% margin on just ₹1.1M. This creates a concentration risk: profit depends on one low-margin category.

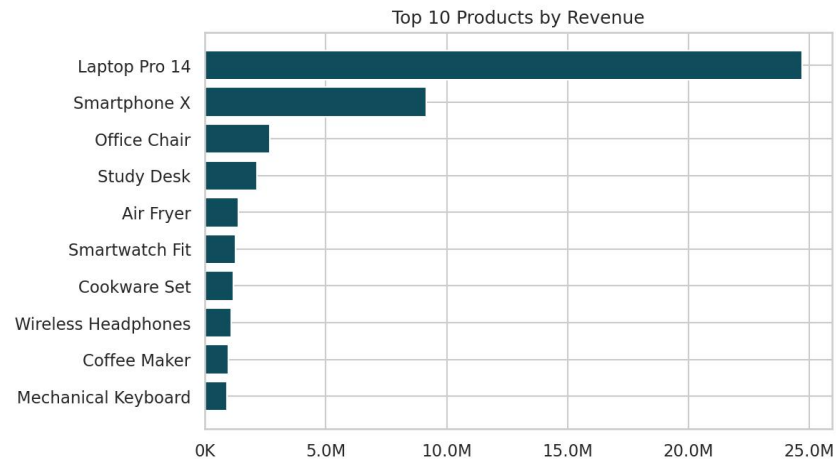


Figure 3: Top 10 products by revenue

Laptop Pro 14 (₹24.7M, 13.7% margin) and Smartphone X (₹9.2M, 12.6% margin) dominate revenue. By profit the top five are Laptop Pro 14, Smartphone X, Office Chair, Study Desk and Air Fryer.

Category-level order outcomes

Non-delivery rates by category: Electronics 9.3%, Home & Kitchen 9.3%, Furniture 8.3%, Sports 7.6%, Fashion 7.5%, Beauty 6.8%.

7. Time Analysis

Year	Revenue	Orders	Margin
2024	₹26.45M	1,478	19.73%
2025	₹26.07M	1,455	20.2%

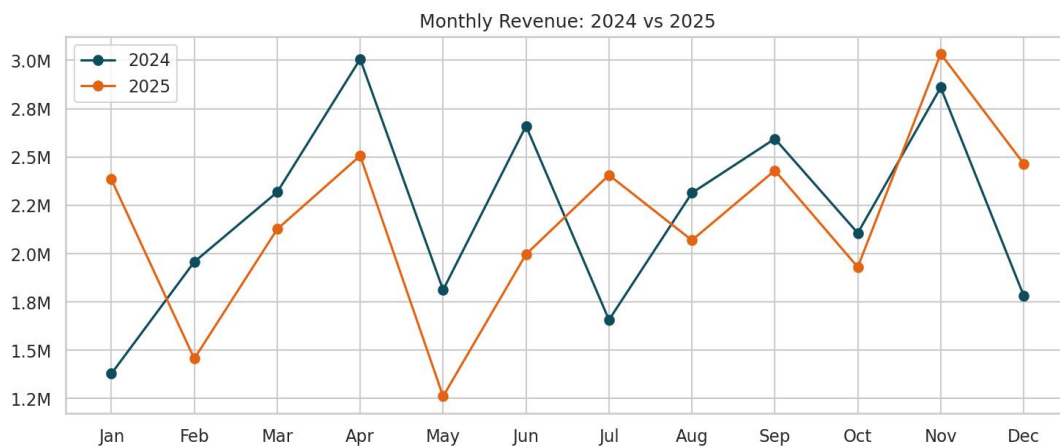


Figure 4: Monthly revenue 2024 vs 2025

- Revenue declined 1.4% in 2025 while margin improved slightly (19.7% → 20.2%).
- Best month: 2025-11 (₹3.03M); worst month: 2025-05 (₹1.26M).
- Q4 is the strongest quarter (27.0% of revenue) and Q1 the weakest (22.1%). May is weak in both years.

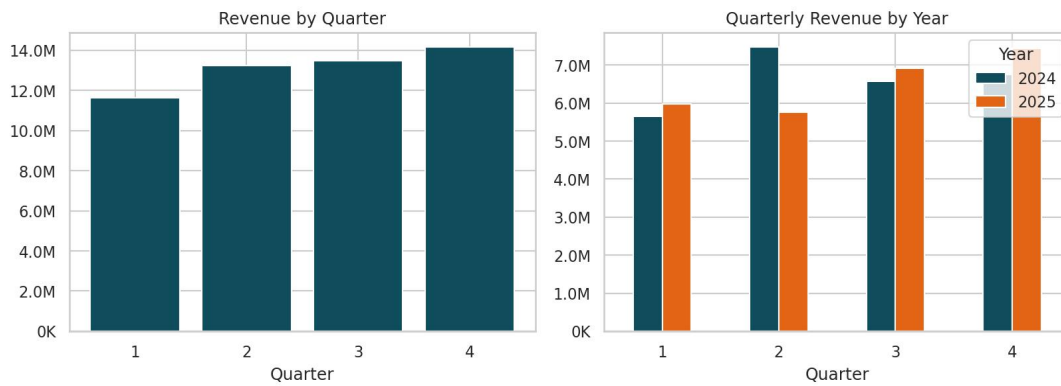


Figure 5: Quarterly revenue

8. Customer Analysis

8.1 Demographics

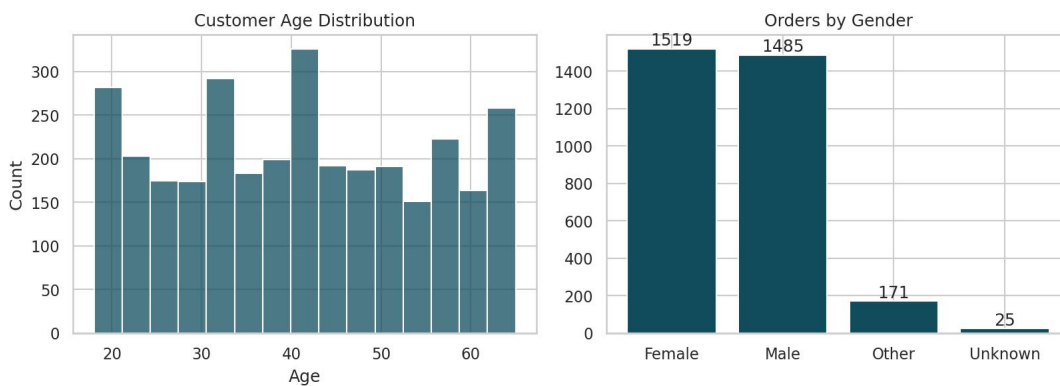


Figure 6: Age distribution and orders by gender

Age is spread evenly from 18 to 65 and every age band contributes roughly ₹10–11M of revenue. Male and female customers place nearly equal numbers of orders (1,374 vs 1,375 delivered); Male customers contribute slightly more revenue (₹25.2M vs ₹23.1M). The Other group has a lower margin (17.1%).

8.2 Purchase behaviour

Customers place on average 3.56 delivered orders (median 3, max 10). 88.3% of customers bought more than once, and the top 20% of customers generate 60.8% of revenue.

8.3 RFM segmentation

Recency (days since last delivered order, reference date = last order date + 1), Frequency (number of orders) and Monetary (total sales) were scored 1–5 by quintile and mapped to rule-based segments.

Segment	Customers	Revenue	Revenue share	Avg orders	Avg recency (days)
Loyal	238	₹16.74M	31.9%	4.4	86.6
Champions	110	₹14.88M	28.3%	5.8	44.3
At Risk	146	₹9.77M	18.6%	4.2	277.2
Lost	238	₹7.06M	13.4%	1.9	328.2

Segment	Customers	Revenue	Revenue share	Avg orders	Avg recency (days)
New Customers	91	₹4.06M	7.7%	2.1	45.8



Figure 7: RFM customer segmentation

Champions (110 customers) and Loyal customers (238) provide 60% of revenue. At Risk customers (146) are frequent buyers whose last purchase was on average 277 days ago and are the best win-back target; 238 Lost customers have low frequency and long inactivity.

9. Geographic Analysis

Region	Revenue	Profit	Orders	Margin
South	₹18.19M	₹3.66M	981	20.14%
West	₹13.96M	₹2.74M	742	19.65%
North	₹11.70M	₹2.39M	679	20.47%
East	₹4.46M	₹0.86M	295	19.21%
Central	₹4.21M	₹0.82M	236	19.61%

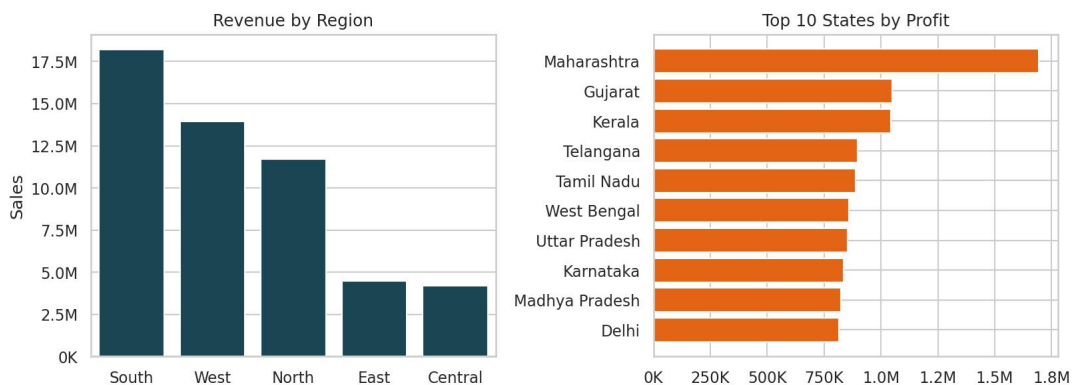


Figure 8: Revenue by region and top 10 states by profit

South generates 34.6% of revenue, followed by West (26.6%) and North (22.3%). Maharashtra is the top state (₹8.2M revenue, ₹1.69M profit), followed by Gujarat and Kerala. Margins are consistent across regions (19.2%–20.5%), so geography drives volume rather than profitability.

10. Sales Channel, Payment & Shipping

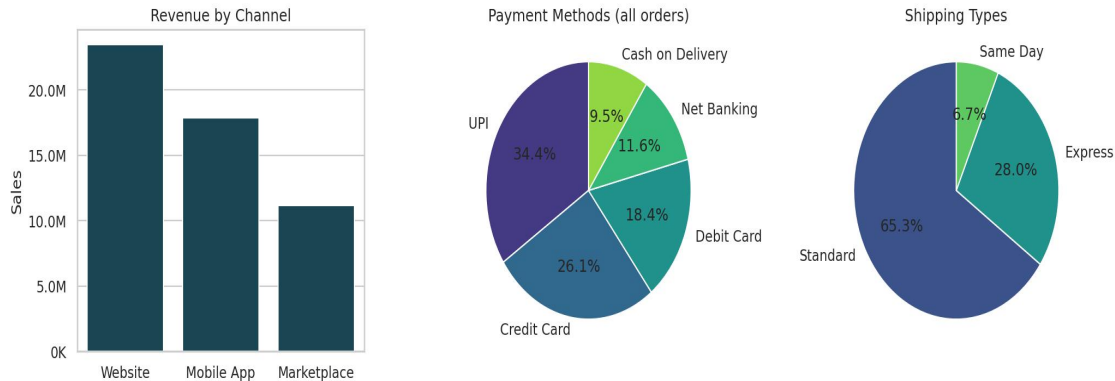


Figure 9: Channel, payment method and shipping type

Sales_Channel	Revenue	Profit	Orders	Margin
Website	₹23.46M	₹4.63M	1331	19.76%
Mobile App	₹17.86M	₹3.54M	1002	19.84%
Marketplace	₹11.19M	₹2.30M	600	20.59%

- Website is the primary channel (44.7% of revenue), then Mobile App (34.0%) and Marketplace (21.3%); margins are nearly identical (19.8%–20.6%).
- UPI is the most popular payment method (34.4% of orders), followed by Credit Card and Debit Card; Cash on Delivery is 9.5% of orders.
- Standard shipping accounts for 65% of orders; Same Day is only 7%. Same Day and Express have slightly higher failure rates (9.8% and 8.8% vs 8.0%).

11. Discount & Profitability

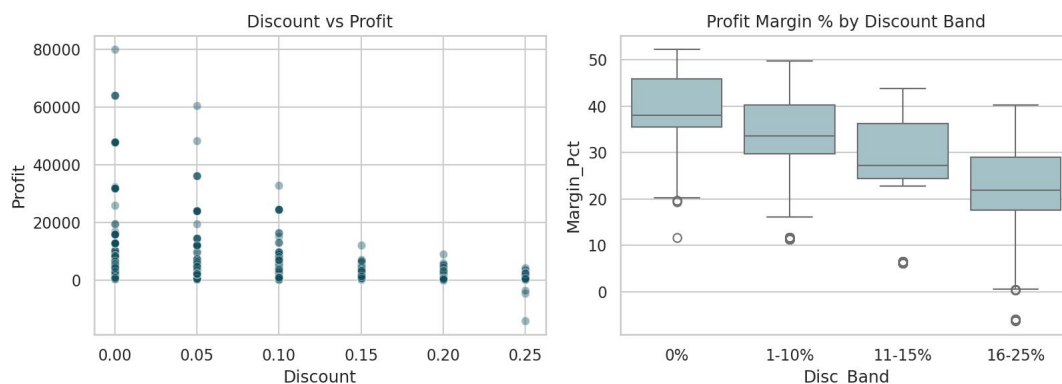


Figure 10: Discount vs profit and margin by discount band

Discount band	Orders	Avg margin
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Discount band	Orders	Avg margin
0%	802	38.6%
1-10%	1416	33.68%
11-15%	518	28.3%
16-25%	464	22.24%

Correlation of discount with margin % is -0.493, with profit -0.241 and with sales -0.064. Discounts therefore reduce margin materially without lifting order value, which suggests they should be targeted rather than blanket.

Correlation and outliers

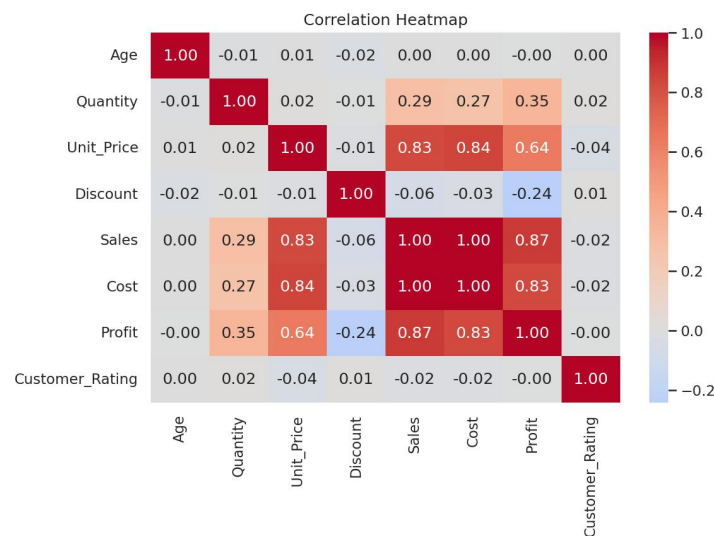


Figure 11: Correlation heatmap

Sales, Cost and Profit are strongly related to Unit_Price (high-ticket electronics). Sales is highly right-skewed: 427 orders (13%) exceed the IQR upper fence, driven by laptops and smartphones. These are genuine high-value orders and were retained.

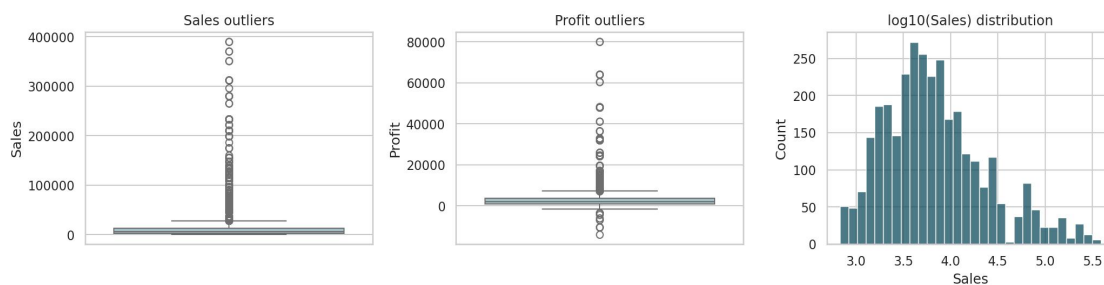


Figure 12: Sales and profit outliers

12. Key Insights

- Business is concentrated: one category (Electronics) and one product (Laptop Pro 14) drive most revenue at the lowest margins.
- Growth has stalled; performance depends on seasonal peaks in Q4.
- Discount depth is the clearest controllable lever on margin.

- Customers are loyal, but nearly half of buyers have lapsed.
- Channel, region and payment method affect volume but hardly change margin.

13. Recommendations

#	Recommendation	Rationale
1	Diversify towards Furniture, Fashion, Beauty and Sports	29–44% margins vs 15% for Electronics
2	Cap blanket discounts near 10%; use targeted offers	Margin drops 16 points from 0% to 16–25% discount
3	Win-back campaign for 146 At Risk customers	₹9.8M lifetime revenue, avg 277 days inactive
4	Reward and retain Champions / Loyal	60% of revenue from 348 customers
5	Investigate returns in Electronics and Home & Kitchen	9.3% non-delivery in both; ₹5.1M gross sales lost
6	Plan stock and campaigns for Q4/November; lift May	Consistent seasonal pattern

14. Limitations & Next Steps

- Two years of data only; seasonality conclusions are indicative.
- Missing discounts assumed 0%; no info on why orders were returned or cancelled.
- Next: sales forecasting (SARIMA/Prophet), churn prediction, pricing optimisation and an interactive dashboard.

Appendix A: Notes on the Original Notebook

The provided notebook was reviewed while preparing this report. Suggested improvements for the final GitHub version:

- Early KPIs and charts use all orders; the dashboard uses delivered orders. Use one consistent basis (this report uses delivered).
- Filling Customer_Rating with the text "No Rating" converts the column to text and breaks mean and correlation calculations; keep NaN instead.
- Ecan_cleaned (dropping null discounts) is created but never used; the discount fill line is commented out.
- Some cells run out of order (top_customers_Ecan used before it is defined), and the dashboard depends on variables defined in earlier cells; use Restart & Run All before publishing.
- Written commentary (e.g. highest revenue months) was based on all orders and will differ from delivered-only results; refresh the text after choosing a basis.
- Data path is a local D: drive path; use a relative path such as data/ecommerce_sales_analytics.xlsx.